

Lab Work 01

Topic	Python Basics, Introduction to OOP, Class and Object
Objective	Practice basic Python syntax and implement the fundamental concepts of classes, objects, attributes, methods, and constructors.
AI Copilot Usage	Not Allowed

Tasks

1. Python Basics Review:

- Write a simple Python function ***calculate_grade(mark)*** that takes a numerical mark (0-100) and returns a letter grade ('HD', 'D', 'C', 'P', 'N') based on standard university grading.
- Test this function with a few different marks.

2. Define the *Student* Class:

- Create a new Python file (e.g., ***school_simulator.py***).
- Define a class named ***Student***.
- Implement the constructor (***__init__***) to accept ***student_id***, ***first_name***, and ***last_name*** as arguments. Store these values as attributes within the object.
- Add an attribute ***units_enrolled***, initialized as an empty list ***[]***.

3. Add Methods to *Student*:

- Add a method ***get_full_name()*** that returns the student's full name (e.g., "FirstName LastName").
- Add a method ***display_details()*** that prints the student's ID and full name to the console.

4. Create and Use *Student* Objects:

- In the main part of your script (outside the class definition, e.g., under ***if __name__ == "__main__":***), create at least two different ***Student*** objects with unique details.
- Call the ***display_details()*** method on each student object.
- Call the ***get_full_name()*** method for one student and print the returned value.
- Access and print the ***units_enrolled*** attribute for one student (it should be empty).

5. Creative Challenge (Optional, if time permits):

- Define a simple ***Unit*** class with attributes like ***unit_code*** and ***unit_name***.
- Add a method ***enroll(self, unit_object)*** to the ***Student*** class that adds a ***Unit*** object to the ***units_enrolled*** list.

- c) Modify ***display_details*** to also list the unit codes the student is enrolled in (if any). Create a ***Unit*** object and test the enrollment.

Important Notes

- This is an individual task. All code must be your own original work.
- Ask questions and seek help from the lecturer or teaching assistants if you are stuck.
- Focus on writing clean, well-organized, and well-documented code.